

Faithfully flat descent

SGA 1, Exposé VIII

English draft

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About this draft

This is an unofficial English translation of Exposé VIII, “Descente fidèlement plate”, from SGA 1. The source volume is A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61. It follows the slightly corrected SMF recomposition, arXiv:math/0206203v2.

This draft contains the whole exposé, including proofs, footnotes, diagrams, the updating remarks added in 2003 by M. Raynaud (between bold brackets and marked (MR)) where the exposé has them, and its bibliography. Statement numbering and mathematical notation follow the source. References to other exposés print the source’s numbers. Apparent misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ VIII

Faithfully flat descent

1. Descent of quasi-coherent Modules

Let **Sch** be the category of preschemes. Proceeding as in VI 11.b, one finds that the category of pairs (X, F) of a prescheme X and a Module F on X (where the morphisms are defined as in loc. cit. with the help of the notion of direct image of a Module by a morphism of ringed spaces) can be regarded as a fibered category over **Sch**, the change of base functor relative to a morphism $f: X \rightarrow Y$ in **Sch** being the inverse image functor of Modules by f . (One will note that the fiber-category at $X \in \text{Ob}(\mathbf{Sch})$ of the preceding fibered category is the category *opposite* to the category of Modules on X .) As the inverse image of a quasi-coherent Module is quasi-coherent, one sees that the full subcategory of the category of pairs (X, F) formed by the pairs for which F is quasi-coherent is a fibered subcategory of the preceding fibered category. (On the other hand, if one makes no hypotheses on f , the direct image of a quasi-coherent Module is not in general a quasi-coherent Module.) One will simply call this fibered category the *fibered category of quasi-coherent Modules on preschemes*.

Let us recall on the other hand that a morphism $f: X \rightarrow Y$ of ringed spaces is called *faithfully flat* if it is *flat* (i.e. for every $x \in X$, $\mathcal{O}_{X,x}$ is a flat module over $\mathcal{O}_{Y,f(x)}$, (IV)) and *surjective*. One says that f is a *quasi-compact* morphism if the inverse image by f of every quasi-compact subset is quasi-compact; when f is a morphism of preschemes, this also means that the inverse image by f of an affine open set of Y is a *finite* union of affine open sets of X .

THEOREM 1.1. *Let $\mathcal{F}$ be the fibered category of quasi-coherent Modules on preschemes. Let $g: S' \rightarrow S$ be a morphism of preschemes, faithfully flat and quasi-compact. Then g is an effective $\mathcal{F}$ -descent morphism.*

Let us recall¹ that this means two things:

COROLLARY 1.2 (Descent of homomorphisms of Modules). *Let $g: S' \rightarrow S$ be a morphism of preschemes, faithfully flat and quasi-compact, F and G two quasi-coherent Modules on S , F' and G' their inverse images on S' , and finally F'' and G'' their inverse images on $S'' = S' \times_S S'$. Consider the diagram of maps of sets defined by the change of base functors by g , p_1 , p_2 (where $p_1, p_2: S' \times_S S' \rightrightarrows S'$ are the two projections):*

$$\text{Hom}_S(F, G) \longrightarrow \text{Hom}_{S'}(F', G') \rightrightarrows \text{Hom}_{S''}(F'', G'').$$

This diagram is exact, i.e. it defines a bijection of the first set onto the set of coincidences of the two maps written from the second into the third.

In other words, the change of base functor by g , $F \mapsto F'$, defines a *fully faithful* functor from the category of quasi-coherent Modules on S into the category of quasi-coherent Modules on S' equipped with a descent datum relative to g . Moreover:

COROLLARY 1.3 (Descent of Modules). *For every quasi-coherent Module F' on S' , every descent datum on F' relative to g is effective, i.e. F' together with its descent datum is*

¹We shall take for granted here the general theory of descent expounded in detail in the article of J. GIRAUD cited in footnote (footnotegiraud) at the foot of the page of the Foreword, a work which we shall cite as [D] in what follows. Cf. also [2] for a succinct exposition.

isomorphic to the inverse image by g of a quasi-coherent Module on S (determined up to unique isomorphism by virtue of 1.2).

In other words, the preceding fully faithful functor is even an *equivalence*. In practice, this means that it comes to the same thing to give oneself a quasi-coherent Module on S , or a quasi-coherent Module on S' equipped with a descent datum relative to g .

Proof of 1.1. First let T be an S -prescheme which is S -isomorphic to the sum of a family of induced open sets S_i of S which cover S . Then it is evident that the structural morphism $T \rightarrow S$ is an effective $\mathcal{F}$ -descent morphism (this means precisely that giving a quasi-coherent Module F on S is equivalent to giving quasi-coherent Modules F_i on the S_i , and gluing isomorphisms $\varphi_{ji}: F_i|_{S_i \cap S_j} \rightarrow F_j|_{S_i \cap S_j}$ satisfying the well-known cochain condition). By virtue of VII, 8 it follows that to verify that $g: S' \rightarrow S$ is an effective $\mathcal{F}$ -descent morphism, it suffices to verify it for the morphism $g_T: T' = T \times_S S' \rightarrow T$ deduced from g by the change of base $T \rightarrow S$. (Note that the hypothesis on $T \rightarrow S$ remains stable under arbitrary change of base, hence that $T \rightarrow S$ is in fact a *universal* effective $\mathcal{F}$ -descent morphism.) Taking for S_i affine open sets which cover S , one is thus reduced to the case where S is affine.

Then S' is a finite union of affine open sets, and taking the S -scheme sum of the latter, one finds an affine S -scheme S_1 and an S -morphism $S_1 \rightarrow S'$ which is flat and surjective. Hence S_1 is also faithfully flat over S . If therefore one proves that a faithfully flat and affine morphism is an effective $\mathcal{F}$ -descent morphism, hence a universal strict $\mathcal{F}$ -descent morphism (the hypothesis being indeed stable under change of base), one concludes in particular that the structural morphism $S_1 \rightarrow S$ is a universal strict $\mathcal{F}$ -descent morphism, and as there exists an S -morphism $S_1 \rightarrow S'$, it will indeed follow, by [D], that $g: S' \rightarrow S$ is a strict $\mathcal{F}$ -descent morphism.

This therefore brings us back to the case where g is an affine morphism, and as we have seen one can then moreover suppose S affine, hence *one can suppose S and S' affine*. In this case, 1.2 is equivalent to the

LEMMA 1.4. *Let A be a ring, A' a faithfully flat A -algebra, M and N two A -modules, M' and N' the A' -modules deduced by change of ring $A \rightarrow A'$, and M'', N'' the $A'' = A' \otimes_A A'$ -modules deduced by change of ring $A \rightarrow A''$. Then the sequence of set-theoretic maps*

$$\mathrm{Hom}_A(M, N) \longrightarrow \mathrm{Hom}_{A'}(M', N') \rightrightarrows \mathrm{Hom}_{A''}(M'', N'')$$

is exact.

As the homomorphism $N \rightarrow N'$ is injective (A' being faithfully flat over A) one sees that the first arrow is injective. It remains to prove that if an A' -homomorphism $u': M' \rightarrow N'$ is compatible with the descent data, then it comes from an A -homomorphism $u: M \rightarrow N$. Now this also means simply that u' maps the subset M of M' into the subset N of N' (the induced map $u: M \rightarrow N$ will then automatically be A -linear since u' is A' -linear, and one sees likewise that u' is necessarily equal to $u \otimes_A A'$). Now if $x \in M$, then $u'(x)$ is an element in the kernel of the pair of maps $N' \rightrightarrows N''$. One is therefore reduced, in order to prove 1.4, to the following particular case (corresponding to the case where $M = A$):

COROLLARY 1.5. *Let N be an A -module; then the sequence of set-theoretic maps*

$$N \longrightarrow N' \rightrightarrows N''$$

is exact.

Indeed, let A_1 be a faithfully flat A -algebra. To show that the sequence under consideration is exact, it suffices to prove that the sequence deduced from it by the change of ring $A \rightarrow A_1$ is so. Now the latter, as one sees at once, is the one relative to the A_1 -module $N_1 = N \otimes_A A_1$ and to the A_1 -algebra $A'_1 = A_1 \otimes_A A'$. It therefore suffices to find an A_1 faithfully flat over A , such that $\mathrm{Spec}(A'_1) \rightarrow \mathrm{Spec}(A_1)$ is a strict $\mathcal{F}$ -descent morphism. Now indeed it suffices to take $A_1 = A'$, for then the preceding morphism admits a right inverse morphism, hence by virtue of [D] it is an effective descent morphism for any fibered category over **Sch**.

It finally remains to show that if N' is an A' -module equipped with a descent datum for $A \rightarrow A'$, i.e. equipped with an isomorphism

$$\varphi: N'_1 \xrightarrow{\sim} N'_2$$

between the two modules deduced from N by the changes of rings $A' \rightrightarrows A' \otimes_A A'$, then N' together with its descent datum is isomorphic to a module $N \otimes_A A'$. One sees easily, taking into account 1.5, that this statement is equivalent to the following:

LEMMA 1.6. *Let N' be an A' -module equipped with a descent datum relative to $A \rightarrow A'$ (where A' is an A -algebra). Let N be the sub- A -module of N' formed by the x such that $\varphi(x \otimes_{A'} 1_{A'}) = 1_{A'} \otimes_{A'} x$, and consider the canonical homomorphism*

$$N \otimes_A A' \longrightarrow N',$$

(which is then compatible with the descent data). If A' is faithfully flat over A , this homomorphism is an isomorphism.

Let us prove this lemma. Let again A_1 be a faithfully flat A -algebra; to show that the morphism under consideration is an isomorphism, it suffices to prove that it becomes one after the change of ring $A \rightarrow A_1$. Now, using the flatness of A_1 over A , one sees that the homomorphism thus obtained is none other than the one that one would obtain directly in terms of the module $N' \otimes_A A_1$ over $A'_1 = A' \otimes_A A_1$, equipped with the descent datum relative to $A_1 \rightarrow A'_1$ which is deduced canonically by change of ring from the one which was given on N' . Thus it suffices to find an A_1 faithfully flat over A such that $\text{Spec}(A'_1) \rightarrow \text{Spec}(A_1)$ is an effective $\mathcal{F}$ -descent morphism. One then takes, as above, $A_1 = A'$. This completes the proof of 1.6, and thereby the proof of 1.1.

COROLLARY 1.7 (Descent of sections of Modules). *Let $g: S' \rightarrow S$ be a morphism of preschemes, faithfully flat and quasi-compact. For every quasi-coherent Module G on S , let G' and G'' be its inverse images on S' and $S'' = S' \times_S S'$, and consider the diagram of homomorphisms of Modules on S :*

$$G \longrightarrow g_*(G') \rightrightarrows h_*(G'')$$

(where $h: S'' \rightarrow S$ is the structural morphism). This diagram is exact.

Indeed, this means that for every open set U in S , the corresponding diagram formed by the sections over U is exact. One can evidently then suppose $U = S$, and the exactness in question is then a particular case of 1.2, obtained by taking $F = \mathcal{O}_S$.

As the inverse image functor of Modules is right exact, one concludes formally from 1.1:

COROLLARY 1.8 (Descent of quotient Modules). *With the notation of 1.7, let moreover, for every quasi-coherent Module F on a prescheme, $\text{Quot}(F)$ be the set of quasi-coherent quotient Modules of F . With this convention, the diagram of maps of sets:*

$$\text{Quot}(G) \longrightarrow \text{Quot}(G') \rightrightarrows \text{Quot}(G'')$$

is exact.

(One would evidently have the same statement with sub-Modules instead of quotient Modules, since the two correspond bijectively.) Taking in particular $G = \mathcal{O}_S$, one finds:

COROLLARY 1.9 (Descent of closed subpreschemes). *For every prescheme X , let $H(X)$ be the set of closed subpreschemes of X . With this notation, and under the conditions of 1.7, the following diagram of maps of sets*

$$H(S) \longrightarrow H(S') \rightrightarrows H(S'')$$

is exact.

It is appropriate to supplement Theorem 1.1 with the following result:

PROPOSITION 1.10 (Descent of properties of Modules). *Let $g: S' \rightarrow S$ be a faithfully flat and quasi-compact morphism, F a quasi-coherent Module on S . For F to be of finite type, resp. of finite presentation, resp. locally free and of finite type, it is necessary and sufficient that its inverse image F' on S' be so.*

One need only prove the “it is sufficient”. One can evidently suppose S affine, and then replacing S' by a sum of affine open sets covering S' , one is reduced to the case where S' is also affine. Then our statement is equivalent to the following:

COROLLARY 1.11. *Let A be a ring, A' a faithfully flat A -algebra, M an A -module, M' the A' -module $M \otimes_A A'$. For M to be of finite type (resp. of finite presentation, resp. locally free of finite type) it is necessary and sufficient that M' be so.*

Indeed, one has $M = \varinjlim_i M_i$, where the M_i are the submodules of finite type of M . Consequently $M' = \varinjlim_i M'_i$, and if M' is of finite type, M' is equal to one of the M'_i , hence by faithful flatness M is equal to M_i , hence M is of finite type. Consequently there exists an exact sequence

$$0 \longrightarrow R \longrightarrow L \longrightarrow M \longrightarrow 0,$$

with L free of finite type, whence an exact sequence

$$0 \longrightarrow R' \longrightarrow L' \longrightarrow M' \longrightarrow 0,$$

with L' free of finite type. If therefore M' is of finite presentation, R' is of finite type, hence by what precedes R is of finite type, hence M is of finite presentation. Finally, to say that M is locally free and of finite type means that it is of finite presentation and flat (cf. IV in the noetherian case; the general case is left to the reader). As each of these properties does descend, the same holds for their conjunction, which completes the proof.

REMARK 1.12. The conjunction of 1.1 and of 1.10 shows that the statement 1.1 still remains valid when one replaces the fibered category $\mathcal{F}$ by the fibered subcategory formed by the quasi-coherent Modules of finite type, resp. of finite presentation, resp. locally free of finite type, resp. locally free of given rank n .

2. Descent of preschemes affine over another one

Since the inverse image functor for Modules is compatible with the tensor product and with other tensor operations, Theorem 1.1 implies various variants, obtained by considering, instead of a single quasi-coherent Module, a quasi-coherent Module or a system of quasi-coherent Modules endowed with various additional structures expressible by means of tensor operations. For example, the datum of three quasi-coherent Modules F, G, H on S and of a pairing

$$F \otimes G \longrightarrow H,$$

is equivalent to the datum of three quasi-coherent Modules F', G', H' on S' , endowed with descent data relative to $g: S' \rightarrow S$, and endowed with a pairing

$$F' \otimes G' \longrightarrow H',$$

“compatible” with these descent data, in the obvious sense of the term. For example, if $F = G = H$, one sees that the datum of a quasi-coherent Module F on S endowed with an algebra law (which for the moment we do not assume to satisfy any axiom of associativity, of commutativity or of existence of a unit section), is equivalent to the same datum on S' , endowed in addition with a descent datum. And using the results of the preceding no., one finds at once that for F to satisfy one of the usual axioms alluded to just now, it is necessary and sufficient that the same hold for F' . For example, the datum of a quasi-coherent Algebra $\mathcal{A}$ on S (by which we henceforth understand: associative, commutative, with unit section) is equivalent to the datum of a quasi-coherent Algebra $\mathcal{A}'$ on S' , endowed with a descent datum relative to $g: S' \rightarrow S$. If one recalls the equivalence between the dual category

of the quasi-coherent Algebras on S , and the category of the S -preschemes affine over S , (EGA II par.1), one at once finds:

THEOREM 2.1. *Let $\mathcal{F}'$ be the fibered category of affine morphisms of preschemes $f: X \rightarrow S$, considered as a fibered subcategory of the fibered category of arrows in the category of preschemes **Sch** (VI 11.a). Let $g: S' \rightarrow S$ be a faithfully flat and quasi-compact morphism of preschemes. Then g is an effective $\mathcal{F}'$ -descent morphism.*

3. Descent of set-theoretic properties and of finiteness properties of morphisms²

PROPOSITION 3.1. *Let $f: X \rightarrow Y$ be an S -morphism, $g: S' \rightarrow S$ a surjective morphism, $f': X' = X \times_S S' \rightarrow Y' = Y \times_S S'$ the morphism deduced from f by the change of base by means of $g: S' \rightarrow S$. For f to be surjective (resp. radicial), it is necessary and sufficient that f' be so.*

We note that f' can also be obtained by the change of base $Y' \rightarrow Y$, which is likewise surjective since it is deduced from $g: S' \rightarrow S$, which is. On the other hand, for every $y \in Y$ and every $y' \in Y'$ over y , we have an isomorphism

$$X'_{y'} \xleftarrow{\sim} X_y \otimes_{k(y)} k(y'),$$

where X_y denotes the fiber of X at y , and $X'_{y'}$ that of X' at y' . It follows that X_y is nonempty (resp. has at most one point, and the latter corresponds to a radicial residue field extension) if and only if $X'_{y'}$ has the same property. This proves 3.1.

COROLLARY 3.2. *Under the conditions of 3.1, if f' is injective (resp. bijective), then f is so as well.*

This comes from the fact that if $X'_{y'}$ has at most one point (resp. exactly one point) the same holds for X_y ; this is indeed so, since the morphism $X'_{y'} \rightarrow X_y$ is surjective (for it is deduced from $\text{Spec}(k(y')) \rightarrow \text{Spec}(k(y))$, which is).

PROPOSITION 3.3. *With the notation of 3.1. Suppose that $g: S' \rightarrow S$ is surjective and quasi-compact (resp. faithfully flat and quasi-compact). For f to be quasi-compact (resp. of finite type), it is necessary and sufficient that f' be so.*

Only the “it suffices” needs to be proved. One may evidently assume $S = Y$, since the hypothesis made on $g: S' \rightarrow S$ is preserved for $Y' \rightarrow Y$. Moreover one may assume Y affine. Then Y' is quasi-compact, hence X' is quasi-compact (since f' is so by hypothesis). Let $(X_i)_{i \in I}$ be a family of affine open sets of X covering X ; then the X'_i are open sets of X' covering X' , hence there is a finite subfamily which covers X' . As $X' \rightarrow X$ is surjective, it follows that the corresponding X_i already cover X , hence X is quasi-compact, i.e. f is quasi-compact. Suppose now f' of finite type, and let us prove that f is so, assuming g faithfully flat. Replacing Y' by the sum of a family of affine open sets covering it, one may assume Y' affine. Finally, X being covered by a finite number of affine open sets X_i by what precedes, one must show that they are of finite type over Y , knowing that X'_i is of finite type over Y' . This then brings us back to the

COROLLARY 3.4. *Let B be an A -algebra, A' a faithfully flat A -algebra, $B' = B \otimes_A A'$ the A -algebra deduced from B by change of ring. For B to be of finite type, it is necessary and sufficient that B' be so.*

Only the “it suffices” needs to be proved. We have $B = \varinjlim_i B_i$, where the B_i are the subalgebras of finite type of B . Hence we have $B' = \varinjlim_i B'_i$, and if B' is of finite type over A' , then B' is equal to one of the B'_i , hence B is equal to B_i , hence is of finite type.

COROLLARY 3.5. *Suppose again that the change of base morphism $g: S' \rightarrow S$ is faithfully flat and quasi-compact. For f to be quasi-finite, it is necessary and sufficient that f' be so.*

²For other results like those treated in nos. 3 and 4, Cf. EGA IV 2.3, 2.6, 2.7.

Indeed, the property “quasi-finite” is by definition the conjunction of “of finite type” and “with finite fibers”, each of which descends well by g , the first by virtue of 3.3, the second by the reasoning of 3.1 (which uses only the fact that g is surjective).

REMARKS 3.6. Let A be a ring, X an A -prescheme. One easily sees that the following conditions are equivalent:

- (i) There exist a noetherian ring A_0 (which one may, if one wishes, assume to be a subring of finite type of A), an A_0 -prescheme of finite type X_0 , a homomorphism $A_0 \rightarrow A$ and an A -isomorphism $X \xrightarrow{\sim} X_0 \times_{A_0} A$.
- (ii) The diagonal morphism $X \rightarrow X \times_{\mathrm{Spec}(A)} X$ is quasi-compact (an empty condition if X is separated over A), X is a finite union of affine open sets X_i whose rings B_i are algebras of finite presentation over A , i.e. quotients of polynomial algebras in a finite number of indeterminates, by ideals of finite type.

If X is itself affine, with ring B , these conditions simply mean that B is an algebra of finite presentation over A .

A morphism $f: X \rightarrow Y$ is called a *morphism of finite presentation*, and one also says that X is of finite presentation over Y , if Y is a union of affine open sets Y_i such that $X|_{Y_i}$, as a Y_i -prescheme, satisfies the preceding equivalent conditions. The same then holds for $X|_{Y'}$ for *every* affine open set Y' in Y . This is a property stable under change of base, and moreover the composite of two morphisms of finite presentation is of finite presentation.

These notions having been set up, one sees on (ii), proceeding as in 1.10, that this statement remains valid when the words “of finite type” are replaced in it by “of finite presentation”.

4. Descent of topological properties

THEOREM 4.1. *Let $g: Y' \rightarrow Y$ be a morphism, and Z a subset of Y . Assume that g is flat, and that there exists a quasi-compact morphism $f: X \rightarrow Y$ such that $Z = f(X)$ (N.B. if Y is noetherian, this last condition is implied by “ Z is constructible”). Then we have*

$$g^{-1}(\overline{Z}) = \overline{g^{-1}(Z)}.$$

One may assume Y affine, then Y' affine. As Y is affine, X is a finite union of affine open sets X_i , and replacing X by the sum of the X_i , one may also assume X affine. Let A, A', B be the rings of Y, Y', X , $B' = B \otimes_A A'$ that of $X' = X \times_Y Y'$, I the kernel of $A \rightarrow B$, I' the kernel of $A' \rightarrow B'$; thus the closed subsets of Y and Y' defined by these ideals are respectively the closure of $Z = f(X)$ and the closure of $Z' = f'(X') = g^{-1}(Z)$. We want to establish that the latter is equal to $g^{-1}(\overline{Z})$, which will follow from $I' = IA'$, itself a consequence of the flatness of A' over A .

COROLLARY 4.2. *Let $g: Y' \rightarrow Y$ be a flat and quasi-compact morphism, and Z' a closed subset of Y' saturated for the set-theoretic equivalence relation defined by g . Then we have $Z' = g^{-1}(g(Z'))$.*

Indeed we have $Z' = g^{-1}(Z)$, with $Z = g(Z')$. One can then apply 4.1, noting that the condition imposed on Z in 4.1 is indeed satisfied by taking for X the prescheme Z' endowed with the reduced structure induced by Y' . (The fact that g is quasi-compact then ensures that the induced morphism $f: Z' \rightarrow Y$ is quasi-compact).

Statement 4.2 also means that *the topology of $g(Y')$ induced by Y is the quotient of that of Y'* . In particular:

COROLLARY 4.3. *Let $g: Y' \rightarrow Y$ be a faithfully flat and quasi-compact morphism. Then g makes Y a quotient topological space of Y' , i.e. for a subset Z of Y , Z is closed (resp. open) if and only if $Z' = g^{-1}(Z)$ is.*

Let us now recall that two elements a, b of Y' have the same image in Y if and only if they are of the form $p_1(c), p_2(c)$ for a suitable element c of $Y'' = Y' \times_Y Y'$. It follows that if g is surjective, we have an *exact* diagram of sets

$$\mathcal{P}(Y) \longrightarrow \mathcal{P}(Y') \rightrightarrows \mathcal{P}(Y''),$$

where for every set E , we denote by $\mathcal{P}(E)$ the set of its subsets. This being so, 4.3 can also be interpreted as follows:

COROLLARY 4.4 (Descent of open resp. closed subsets). *Let $g: Y' \rightarrow Y$ be as in 4.3. For every prescheme X , let $\text{Ouv}(X)$ resp. $\text{Fer}(X)$ be the set of its open subsets resp. the set of its closed subsets. Then we have exact diagrams of maps of sets (deduced from g and from the two projections of $Y'' = Y' \times_Y Y'$):*

$$\text{Ouv}(Y) \longrightarrow \text{Ouv}(Y') \rightrightarrows \text{Ouv}(Y'')$$

$$\text{Fer}(Y) \longrightarrow \text{Fer}(Y') \rightrightarrows \text{Fer}(Y'')$$

We have the following complement to 4.3:

COROLLARY 4.5. *Let $g: Y' \rightarrow Y$ be as in 4.3, and let Z be a subset of Y such that there exists a quasi-compact morphism $f: X \rightarrow Y$ with image Z (for example Z constructible, Y noetherian). For Z to be a locally closed subset of Y , it is necessary and sufficient that $Z' = g^{-1}(Z)$ be a locally closed subset of Y' .*

It suffices to prove the “it suffices”. Let Y_1 be the closed subscheme of Y , closure of Z endowed with the induced reduced structure, and let $Y'_1 = Y_1 \times_Y Y'$ be the closed subscheme of Y' inverse image of Y_1 . Its underlying set is the inverse image $g^{-1}(Y_1) = g^{-1}(\overline{Z})$, hence is equal, by virtue of 4.1, to $\overline{Z'}$. As Z' is locally closed in Y' , it is open in $\overline{Z'}$, hence open in Y'_1 . Now the latter is faithfully flat and quasi-compact over Y_1 , hence by virtue of 4.3 one concludes that Z is open in Y_1 , i.e. in $\overline{Z}$, which means that Z is locally closed.

COROLLARY 4.6. *Let $g: S' \rightarrow S$ be a faithfully flat and quasi-compact morphism, $f: X \rightarrow Y$ an S -morphism, $f': X' \rightarrow Y'$ the S' -morphism deduced from it by change of base. Suppose that f' is an open map (resp. a closed map, resp. quasi-compact and a homeomorphism into, resp. a homeomorphism onto); then f has the same property.*

As Y' is faithfully flat and quasi-compact over Y , one may assume $Y = S$. Let Q be a subset of X ; we then have (denoting by h the projection morphism $X' \rightarrow X$):

$$g^{-1}(f(Q)) = f'(h^{-1}(Q)).$$

If Q is open (resp. closed), the same holds for $h^{-1}(Q)$, hence also for $f'(h^{-1}(Q))$ if f' is assumed to be an open map (resp. a closed map), hence the same holds for $f(Q)$ by virtue of the preceding formula and of 4.3. This proves the first two assertions in 4.6; it remains to examine the case where f' is a homeomorphism into, and to prove then that f is a homeomorphism into. (The case of a homeomorphism onto will then follow from 3.1). By virtue of 3.2 f is injective; it remains to prove that the map $X \rightarrow f(X)$ is open. We already know that f is quasi-compact by virtue of 3.3. It suffices to prove that for every closed subset Z of X , we have $Z = f^{-1}(\overline{f(Z)})$, which is equivalent to the analogous formula for the inverse images under the surjective map $h: X' \rightarrow X$, i.e. to

$$Z' = f'^{-1}(g^{-1}(\overline{f(Z)})),$$

where we set $Z' = h^{-1}(Z)$. Now by virtue of 4.1 applied to the subset $f(Z)$ of Y , we have $g^{-1}(\overline{f(Z)}) = \overline{g^{-1}(f(Z))}$, and the formula to be proved is equivalent to

$$Z' = f'^{-1}(\overline{f'(Z')}),$$

which follows from the hypothesis that f' is a homeomorphism into.

N.B. In this last argument, having already assumed that f is quasi-compact, we did not use that g is quasi-compact, but only that g is faithfully flat. Hence it is under this hypothesis

that one can descend the property “homeomorphism into”, or “homeomorphism onto”, or also, thanks to the preceding argument, the property “ f' is quasi-compact and makes $f'(X')$ a quotient topological space of X' ”.

We shall say that a morphism $f: X \rightarrow Y$ of preschemes is *universally open* (resp. *universally closed*, resp. *universally bicontinuous*, etc.) if for every change of base $Y' \rightarrow Y$, $f': X' \rightarrow Y'$ is an open morphism (resp. closed, resp. a homeomorphism onto the image space). One then derives from 4.6:

COROLLARY 4.7. *Under the conditions of 4.6, for f to be universally open, (resp. universally closed, resp. a universal homeomorphism into, resp. a universal homeomorphism), it is necessary and sufficient that f' be so.*

COROLLARY 4.8. *Under the conditions of 4.6, for f to be separated (resp. proper) it is necessary and sufficient that f' be so.*

To say that f is separated means that the diagonal morphism $X \rightarrow X \times_Y X$ is closed, or also universally closed, and the first assertion of 4.8 therefore follows from 4.7. To say that f is proper means that f satisfies the conditions a) f is of finite type b) f is separated c) f is universally closed. Condition a) descends well by virtue of 3.3, b) also according to what we have just seen, and finally c) likewise by 4.7.

REMARKS 4.9. Let us recall that when $g: Y' \rightarrow Y$ is a flat morphism of finite type, with Y locally noetherian, then g is an open morphism (VI 6.6), which is a more precise result than 4.3. One will note however that if f is a faithfully flat and quasi-compact morphism of noetherian preschemes, then f is not in general an open morphism. Let for example Y be an irreducible scheme whose generic point y is not open (for example an algebraic curve), and take for Y' the sum scheme $Y \amalg \text{Spec}(k(y))$; then the image under the structural morphism $Y' \rightarrow Y$ of the open subset $\text{Spec}(k(y))$ is not an open subset of Y . The reader will also notice that various statements of the present exposé become false if one drops from them the hypothesis that the faithfully flat morphism considered is also quasi-compact, the typical case in which the statements fail being that where one takes for Y' the sum scheme of the spectra of the local rings of the points of Y . For example, taking again for Y an irreducible algebraic curve, and for Z the subset of Y reduced to the generic point, its inverse image in Y' is open, without Z being open.

4.10. Various statements of the present exposé remain valid when the hypothesis that Y' is flat over Y is replaced in them by the following: there exists a Module of finite type F on Y' , with support Y' , flat with respect to Y ; the hypothesis of faithful flatness will then be replaced by the preceding one, plus the hypothesis that $Y' \rightarrow Y$ is surjective. This applies to the first two assertions in 1.10, to 3.3, 3.5, 4.1, and consequently to all the results of the present no.

5. Descent of morphisms of preschemes

PROPOSITION 5.1. *Let $g: S' \rightarrow S$ be a morphism of preschemes.*

a) *Suppose that g is surjective, and that the homomorphism*

$$g^*: \mathcal{O}_S \longrightarrow g_*(\mathcal{O}_{S'})$$

is injective; then g is an epimorphism in the category of preschemes, and even in the category of ringed spaces.

b) *Suppose that g is surjective and makes S a topological quotient space of S' . Let $S'' = S' \times_S S'$ and let $h: S'' \rightarrow S$ be the structural morphism; consider the canonical diagram of homomorphisms:*

$$\mathcal{O}_S \longrightarrow g_*(\mathcal{O}_{S'}) \rightrightarrows h_*(\mathcal{O}_{S''}) ,$$

and suppose this diagram exact. Then g is an effective epimorphism in the category of preschemes (and also in the category of ringed spaces), i.e. the diagram

$$S \longleftarrow S' \rightrightarrows S''$$

is exact.

Proof. a) One must show that a morphism of ringed spaces $f: S \rightarrow Z$ is known when one knows fg . Now since g is surjective, one knows the set-theoretic map f_0 underlying f ; it remains to determine the homomorphism of sheaves of rings $\mathcal{O}_Z \rightarrow \mathcal{O}_S$, or what amounts to the same, the homomorphism of sheaves of rings

$$u: f_0^{-1}(\mathcal{O}_Z) \longrightarrow \mathcal{O}_S$$

defined by f . One already knows the homomorphism

$$(fg)_0^{-1}(\mathcal{O}_Z) = g_0^{-1}(f_0^{-1}(\mathcal{O}_Z)) \longrightarrow \mathcal{O}_{S'}$$

defined by fg , or what amounts to the same, one has at one's disposal a homomorphism

$$f_0^{-1}(\mathcal{O}_Z) \longrightarrow g_{0*}(\mathcal{O}_{S'}) = g_*(\mathcal{O}_{S'}).$$

One observes at once that the latter is nothing but the composite of $g^*: \mathcal{O}_S \rightarrow g_*(\mathcal{O}_{S'})$ and of u , and since g^* is injective, u is known when one knows g^*u . [N.B. we have of course not used the fact that $g: S' \rightarrow S$ is a morphism of preschemes; the statement would hold for an arbitrary morphism of ringed spaces; the same remark applies to b), both in the category of ringed spaces and in the category of spaces ringed in local rings. Note also that if g is a morphism of preschemes that is not necessarily surjective, but such that $g^*: \mathcal{O}_S \rightarrow g_*(\mathcal{O}_{S'})$ is injective, then for two morphisms f_1, f_2 from S into a *scheme* Z such that $f_1g = f_2g$, one has $f_1 = f_2$; indeed, if I is the Ideal on S which defines the subprescheme of S of coincidences of f_1, f_2 (inverse image of the diagonal subprescheme of $Z \times Z$ under (f_1, f_2)), one sees that I is contained in $\text{Ker}(g^*)$].

b) One must show that for every ringed space Z , the following diagram of maps

$$\text{Hom}(S, Z) \longrightarrow \text{Hom}(S', Z) \rightrightarrows \text{Hom}(S'', Z)$$

is exact, and that the same holds when Z is a space ringed in local rings and one restricts oneself to homomorphisms of spaces ringed in local rings. Since one already knows by a) that the first map is injective, it remains to see that if $f': S' \rightarrow Z$ is a homomorphism of ringed spaces such that $f'p_1 = f'p_2$, then f' is of the form fg , where $f: S \rightarrow Z$ is a homomorphism of ringed spaces. Since g is surjective, it is then evident that if f' is a morphism of spaces ringed in local rings, the same will hold for f .

From the hypothesis on f' it follows that the underlying set-theoretic map f'_0 is constant on the fibers of the map g_0 , hence, since the latter is surjective, f'_0 factors uniquely as $f'_0 = f_0g_0$, where $f_0: S \rightarrow Z$ is a map, necessarily continuous since g_0 identifies S with a topological quotient space of S' . Now consider the homomorphism

$$f_0^{-1}(\mathcal{O}_Z) \longrightarrow g_*(\mathcal{O}_{S'})$$

deduced from the homomorphism $(f_0g_0)^{-1}(\mathcal{O}_Z) \rightarrow \mathcal{O}_{S'}$ corresponding to f' . The hypothesis $f'p_1 = f'p_2$ is then interpreted as saying that the composites of the preceding homomorphism with the two homomorphisms $g_*(\mathcal{O}_{S'}) \rightrightarrows h_*(\mathcal{O}_{S''})$ are the same, hence by hypothesis b) it factors through a morphism

$$f_0^{-1}(\mathcal{O}_Z) \longrightarrow \mathcal{O}_S.$$

The latter defines a morphism of ringed spaces $f: S \rightarrow Z$, which is the desired morphism.

THEOREM 5.2. *Let $\mathcal{F}$ be the fibered category of arrows in the category **Sch** of preschemes (VI 11.a). Then every faithfully flat and quasi-compact morphism $g: S' \rightarrow S$ is an $\mathcal{F}$ -descent morphism (or again, as one says, a descent morphism in **Sch**).*

This therefore means the following: let $S'' = S' \times_S S'$, and for two preschemes X, Y over S , consider their inverse images X', Y' over S' and their inverse images X'', Y'' over S'' , whence a diagram of maps

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_{S'}(X', Y') \rightrightarrows \mathrm{Hom}_{S''}(X'', Y'') ;$$

with these notations, 5.2 means that this diagram is exact. Note that it is not true in general that g is an effective descent morphism, i.e. that for every prescheme X' over S' , every descent datum on X' relative to $g: S' \rightarrow S$ is effective. The question of effectiveness, often delicate, will be examined in no. 7.

We have seen [D], (taking into account that fibered products exist in **Sch**) that statement 5.2 is equivalent to the following:

COROLLARY 5.3. *A faithfully flat and quasi-compact morphism of preschemes is a universal effective epimorphism.*

Since a faithfully flat and quasi-compact morphism remains such under every extension of the base, one is reduced to proving that it is an effective epimorphism. One then applies criterion 5.1 b), which gives the desired result, taking into account 4.3 and 1.7.

COROLLARY 5.4. *Let $g: S' \rightarrow S$ be a faithfully flat and quasi-compact morphism, $f: X \rightarrow Y$ an S -morphism, $f': X' \rightarrow Y'$ the S' -morphism deduced from it by the change of base $S' \rightarrow S$. For f to be an isomorphism, it is necessary and sufficient that f' be one.*

Indeed, if f' is an isomorphism, it is also an isomorphism for the natural descent structures on X', Y' , and since the functor $X \mapsto X'$ from **Sch**/ S into the category of preschemes over S' endowed with a descent datum relative to g is fully faithful by 5.2, the desired conclusion follows.

COROLLARY 5.5. *Under the conditions of 5.4, for f to be a closed immersion (resp. an open immersion, resp. a quasi-compact immersion) it is necessary and sufficient that f' be one.*

One may suppose as usual that $Y = S$, and one has to prove only the “it is sufficient”. Note that the fact that X'/Y' is endowed with a descent datum relative to $g: Y' \rightarrow Y$, and that the structural morphism $f': X' \rightarrow Y'$ is an immersion, hence a monomorphism, implies that the two subobjects of Y'' that are the inverse images of X'/Y' under the one and the other projection of S'' into S' are the same. If f' is a closed immersion, it follows by virtue of 1.9 that there exists a closed subprescheme X_1 of Y whose inverse image under $g: Y' \rightarrow Y$ is X' . Hence by the uniqueness of the solution of a descent problem relative to an $\mathcal{F}$ -descent morphism, it follows that X_1 is Y -isomorphic to X , hence $f: X \rightarrow Y$ is a closed immersion. One proceeds in the same way for an open immersion, using 4.4. If finally f' is a quasi-compact immersion, f is quasi-compact by virtue of 3.3, hence one can apply to the subset $f(X)$ of Y the criterion 4.5, which proves that $f(X)$ is locally closed since its inverse image $f'(X')$ in Y' is so. Replacing then Y by an open subset in which $f(X)$ is closed, one is reduced to the case where f' is a closed immersion, hence f is one by virtue of what precedes.

COROLLARY 5.6. *Under the conditions of 5.4, for f to be affine, it is necessary and sufficient that f' be so.*

One proceeds as in 5.5, using 2.1 (One can also use Serre’s cohomological criterion [EGA II 5.2], which proves 5.6 without using any descent technique).

COROLLARY 5.7. *Under the conditions of 5.4, for f to be integral (resp. finite, resp. finite and locally free) it is necessary and sufficient that f' be so.*

One has to prove only the “it is sufficient”, and as usual one may suppose $Y = S$, Y affine, and Y' affine. Since the hypothesis implies that f' is affine, the same holds for f by 5.6, hence X and consequently X' are affine. Let $A, A', B, B' = B \otimes_A A'$ be the rings

of Y, Y', X, X' . One has $B = \varinjlim_i B_i$, where B_i runs through the sub- A -algebras of B that are of finite type over A , whence $B' = \varinjlim_i B'_i$, where the B'_i are subalgebras of finite type of the A' -algebra B' . If B' is integral over A' , the B'_i are modules of finite type over A' , hence, A' being faithfully flat over A , the B_i are modules of finite type over A , i.e. B is integral over A . One sees in the same way that if B' is finite over A' , B is so over A . Same conclusion for “locally free of finite type”, cf. 1.11.

COROLLARY 5.8. *Under the conditions of 5.4, suppose f quasi-compact and let $\mathcal{L}$ be an invertible Module on X , and $\mathcal{L}'$ its inverse image on X' . For $\mathcal{L}$ to be ample (resp. very ample) relative to f , it is necessary and sufficient that $\mathcal{L}'$ be ample (resp. very ample) relative to f' .*

One has to prove only the “it is sufficient”. The hypothesis on $\mathcal{L}$ implies in any case that f' is separated, hence f is separated by 4.8, and since f is quasi-compact, and $g: Y' \rightarrow Y$ is flat, the computation of direct images by affine coverings shows that one has isomorphisms

$$g^*(f_*(\mathcal{L}^{\otimes n})) \xrightarrow{\sim} f'_*(\mathcal{L}'^{\otimes n})$$

for every integer n , hence one has an isomorphism

$$g^*(\mathcal{S}) \xrightarrow{\sim} \mathcal{S}',$$

where $\mathcal{S}$ (resp. $\mathcal{S}'$) denotes the quasi-coherent graded Algebra on Y (resp. on Y') that is the direct sum of the $f_*(\mathcal{L}^{\otimes n})$ (resp. of the $f'_*(\mathcal{L}'^{\otimes n})$) for $n \geq 0$. Note that for every $n \geq 0$, the cokernel of the canonical homomorphism $f'^*(\mathcal{S}'_n) \rightarrow \mathcal{L}'^{\otimes n}$ is the inverse image under $X' \rightarrow X$ of the cokernel of $f^*(\mathcal{S}_n) \rightarrow \mathcal{L}^{\otimes n}$, hence its support Z'_n is the inverse image of the support Z_n . If $\mathcal{L}'$ is ample the intersection of the Z'_n is empty, hence, since $X' \rightarrow X$ is surjective, the intersection of the Z_n is empty, i.e. one has a canonical morphism

$$j: X \rightarrow \text{Proj}(\mathcal{S})$$

(EGA II 3). Moreover, the analogous morphism

$$j': X' \rightarrow \text{Proj}(\mathcal{S}')$$

is none other than the one deduced from the preceding one by the change of base $Y' \rightarrow Y$ (loc. cit.). This being so, to say that $\mathcal{L}'$ is ample relative to f' means that j' is an immersion, necessarily quasi-compact moreover, since f' is quasi-compact. Hence by virtue of 5.5 j is an immersion, i.e. $\mathcal{L}$ is ample relative to f . — One proceeds in an entirely analogous way in the case of “very ample”, restricting oneself above to $n = 1$, and replacing the consideration of $\text{Proj}(\mathcal{S})$ by that of the projective bundle $\mathcal{P}(\mathcal{S}_1)$ associated to $\mathcal{S}_1$.

Recall (EGA II 5.1.1) that a quasi-compact morphism f is said to be *quasi-affine* if for every affine open set U in Y , $f^{-1}(U)$ is a prescheme isomorphic to an open subscheme of an affine scheme. One shows (loc. cit.) that it amounts to the same to say that $\mathcal{O}_X$ is ample (or also: very ample) relative to f . Hence 5.8 implies:

COROLLARY 5.9. *Under the conditions of 5.4, and supposing f quasi-compact, for f to be quasi-affine, it is necessary and sufficient that f' be so.*

REMARKS 5.10. Hironaka’s example of a non-projective variety shows that one can have a proper morphism $f: X \rightarrow Y$ of nonsingular algebraic varieties (with Y projective), such that Y is the union of two open sets Y_i such that $X_i = X \times_Y Y_i$ is projective over Y_i , without f being projective. Hence, setting $Y' = Y_1 \amalg Y_2$, Y' is faithfully flat and quasi-compact (and even quasi-finite) over Y , $f': X' \rightarrow Y'$ is projective, but f is not projective. One must therefore be careful that in order to apply 5.8, and to deduce from the fact that f' is projective the same conclusion for f , one must already have on X' an invertible Module $\mathcal{L}'$ ample for f' , endowed with a descent datum relative to $X' \rightarrow X$, (which allows one to regard $\mathcal{L}'$ as the inverse image of an invertible Module $\mathcal{L}$ on X , which will then be ample for f thanks to 5.8). When $g: S' \rightarrow S$ is finite and locally free, see however 7.7.

6. Application to finite and quasi-finite morphisms³

We are going to prove the following two theorems:

THEOREM 6.1. *Let $f: X \rightarrow Y$ be a proper morphism with finite fibers, with Y locally noetherian. Then f is finite.*

THEOREM 6.2. *Let $f: X \rightarrow Y$ be a quasi-finite and separated morphism, with Y locally noetherian. Then f is quasi-affine, and a fortiori quasi-projective.*

REMARKS 6.3. Theorem 6.1 is well known, and due to Chevalley in the case of algebraic varieties; one will also find a simple proof of it in [EGA III 4], using the “theorem on holomorphic functions”. The proof given here does not use this latter theorem, but on the other hand uses the theory of descent; we give it as a bonus for the reader, since one gets it “for nothing” at the same time as that of 6.2. Recall also ([EGA III 4] or [1]) that the global form of Zariski’s “Main Theorem”, deduced from the “theorem on holomorphic functions”, asserts that if $f: X \rightarrow Y$ is quasi-finite and *quasi-projective*, Y being noetherian, then X is Y -isomorphic to an open subprescheme of a *finite* Y -prescheme Z . The conjunction of the “Main Theorem” and of 6.2 can therefore be stated as follows:

COROLLARY 6.4. *Let $f: X \rightarrow Y$ be a quasi-finite and separated morphism, with Y noetherian. Then X is Y -isomorphic to an open subprescheme of a finite Y -prescheme Z .*

Another interesting consequence of 6.2 for the theory of descent will be given with 7.9.

Proof of 6.1 and 6.2. We shall admit the following fact, whose proof is easy⁴:

LEMMA 6.5. *Let X be a prescheme of finite type over Y locally noetherian, and let $y \in Y$. For there to exist an open neighborhood U of y such that $X|U$ is finite (resp. quasi-affine, resp. ...) over U , it is necessary and sufficient that $X \times_Y \text{Spec}(\mathcal{O}_y)$ be finite (resp. quasi-affine, resp. ...) over $\text{Spec}(\mathcal{O}_y)$.*

Since on the other hand the property for $f: X \rightarrow Y$ of being finite, resp. quasi-affine, is local on Y , one is reduced, in order to prove 6.1 and 6.2, to the case where Y is the spectrum of a local ring, and is a fortiori of finite dimension. (N.B. one calls *dimension of a prescheme* Y the sup of the Krull dimensions of its local rings). We proceed by induction on

$$n = \dim(Y),$$

the assertion being trivial for $n < 0$. We may therefore suppose $n \geq 0$, and the assertion proved for the dimensions $n' < n$. One may again suppose that Y is the spectrum of a noetherian local ring A , of dimension n . Note that the hypotheses made in 6.1 and 6.2 are stable under change of base (we have already used this in the reduction at the beginning), so they will remain true after the change of base $\text{Spec}(\hat{A}) \rightarrow \text{Spec}(A)$. Since the latter is faithfully flat and quasi-compact, statements 5.7 and 5.9 reduce us to the case where A is moreover complete. Using then the fact that every noetherian local ring B over A that is quasi-finite over A is finite over A , and the fact that X is separated over Y and the fiber of y consists of isolated points, one finds a decomposition

$$X = X' \amalg X''$$

where X' is *finite* over Y , and where the fiber of X'' at y is empty. If X is proper over Y , so is X'' , hence its image in Y is closed, and since it does not contain y , it is empty, hence $X'' = \emptyset$, hence $X = X'$, which shows that X is finite over Y and proves 6.1 (N.B. the induction hypothesis is useless here). If X is quasi-finite over Y , so is X'' ; now X'' in fact lies over the open set $Y - \{y\}$ of Y , which is of dimension $< n$. By virtue of the induction hypothesis, X'' is quasi-affine over $Y - \{y\}$, hence also over Y ; the same obviously holds for X' , hence also for their sum X , which proves 6.2.

³Cf. EGA IV 18.12 for generalizations to preschemes that are not necessarily locally noetherian

⁴Cf. EGA IV 8

REMARK 6.6. Theorems 6.1 and 6.2 remain valid if one no longer supposes Y locally noetherian, provided one specifies that f is supposed to be of finite presentation (cf. 3.6). Indeed, one can still suppose Y affine, and then one verifies without difficulty that the situation $f: X \rightarrow Y$ is deduced, by a change of base $Y \rightarrow Y_0$, from a situation $f_0: X_0 \rightarrow Y_0$ satisfying the same hypotheses as f , with Y_0 *noetherian*. Hence by result 6.1 resp. 6.2, f_0 is finite resp. quasi-affine, hence so is f . This kind of reasoning is often useful for getting rid of noetherian hypotheses, (which always end up being troublesome in the applications).

7. Effectiveness criteria for a descent datum

Consider as usual a morphism of preschemes

$$g: S' \longrightarrow S$$

and an S' -prescheme X' . In accordance with the general facts VII, 9⁵, giving a descent datum on X' , relative to g , is equivalent to giving an equivalence pair [3]:

$$q_1, q_2: X'' \rightrightarrows X'$$

such that the structural morphism $X' \rightarrow S'$ is compatible with this pair and the equivalence pair

$$p_1, p_2: S'' = S' \times_S S' \rightrightarrows S'$$

defined by g , and such that the two squares (or one of the two, which amounts to the same by reason of symmetry) extracted from the corresponding diagram

$$\begin{array}{ccc} X' & \xleftarrow{\quad} & X'' \\ \downarrow & & \downarrow \\ S' & \xleftarrow{\quad} & S'' \end{array}$$

(using either p_1, q_1 , or p_2, q_2), are *cartesian*. A solution of the descent problem posed by this descent datum, i.e. an object X over S endowed with an isomorphism $X \times_S S' \xleftarrow{\sim} X'$ compatible with the descent data, is equivalent to giving a *cartesian* square

$$\begin{array}{ccc} X & \xleftarrow{h} & X' \\ \downarrow & & \downarrow \\ S & \xleftarrow{g} & S' \end{array}$$

satisfying $hq_1 = hq_2$.

Since the set of faithfully flat and quasi-compact morphisms is stable under change of base, and since a faithfully flat and quasi-compact morphism is an effective epimorphism by virtue of 5.3, it follows from the general theory [D]:

PROPOSITION 7.1. *Suppose $g: S' \rightarrow S$ faithfully flat and quasi-compact. For a descent datum on X' relative to g to be effective, it is necessary and sufficient that the equivalence relation $R = (q_1, q_2)$ which it defines be effective (i.e. the quotient X'/R exists and X'' becomes the fibered square of X' over X'/R), and that the canonical morphism $X' \rightarrow X'/R$ be faithfully flat and quasi-compact.*

Thus the question of the effectiveness of a descent datum is a special case of the question of effectiveness of an equivalence graph, and various effectiveness criteria given in this no. can be obtained in this way. Nevertheless, in the context of descent one has at one's disposal theorem 2.1, which implies that *if X' is affine over S' , every descent datum on X' relative to g is effective*, a statement which has no analogue for passage to the quotient by a general flat equivalence graph. All the effectiveness criteria that we give here can also be considered as deduced from the preceding one.

⁵does not exist; see exposé 212 of the Séminaire Bourbaki (1962)

Let U' be a subscheme of X' (or more generally a subobject of X' in the category **Sch**); one says that U' is *stable under the descent datum* on X' if one can find on U' a descent datum relative to g such that the immersion $U' \rightarrow X'$ is compatible with the descent data. This also means that the inverse images of U' in X'' under q_1 and q_2 are the same (or also, as one says, that U' is *stable under the equivalence relation* R), and of course the descent datum in question on U' is then unique, and is called the *descent datum induced* by that of X' . This being said:

PROPOSITION 7.2. *Let (X'_i) be a covering of X' by open sets X'_i stable under the descent datum. For the descent datum on X' to be effective, it is necessary and sufficient that the same hold for the induced descent data on the X'_i .*

This is an easy consequence of 7.1 for instance, and the details of the proof are left to the reader.

COROLLARY 7.3. *Let (S_i) be an open covering of S , and for every i let S'_i and X'_i be deduced from S' and X' by the change of base $S_i \rightarrow S$. For the descent datum on X' to be effective, it is necessary and sufficient that, for every i , the descent datum on X'_i relative to $g_i: S'_i \rightarrow S_i$ be effective.*

This criterion practically always reduces us to the case where S is affine. In the case where S' is also affine, which is the most frequent case in applications, one has:

COROLLARY 7.4. *Suppose S and S' affine. For the descent datum on X' to be effective, it is necessary and sufficient that X' be a union of open sets X'_i that are affine and stable under the descent datum.*

The sufficiency comes from 7.2 and from the fact that if X'_i is affine, it is affine over S' and one can apply 2.1. For the necessity, one notes that if X' comes from X , and if X is covered by affine open sets X_i , then the $X'_i = X_i \times_S S'$ are affine open sets stable under the descent data and covering X' .

COROLLARY 7.5. *Let $g: S' \rightarrow S$ be a faithfully flat, quasi-compact and radicial morphism. Then g is an effective descent morphism, i.e. for every X' over S' , every descent datum on X' relative to $g: S' \rightarrow S$ is effective.*

Indeed, by virtue of 7.3 one may assume S affine; hence, as S' is radicial over S and hence separated, S' is separated. Moreover, for every $x' \in X'$, the fiber $R(x') = q_2(q_1^{-1}(x'))$ of the set-theoretic equivalence relation defined by the equivalence relation R is reduced to a point, because, g being radicial, the same holds for p_1, p_2 , which are deduced from it by the change of base $S' \rightarrow S$, hence also for q_1, q_2 , which are deduced from the preceding ones by the change of base $X' \rightarrow S''$. Hence *every open set of X' is stable* under the descent datum. Let us then cover X' by affine open sets X'_i ; they are affine over S since S' is separated, hence the induced descent datum is effective by 2.1. One then concludes by 7.2.

One will note that 7.5 gives the only known case of an effective descent morphism in the category of preschemes, and it is probably indeed the only case, even when restricting oneself to noetherian schemes, or to schemes of finite type over a field.

When one assumes S locally noetherian and S' of finite type over S , the statement 7.5 is also a special case of the following one (which generalizes Weil's Galois descent and Cartier's inseparable descent):

COROLLARY 7.6. *Let $g: S' \rightarrow S$ be a finite locally free morphism (i.e. defined by an Algebra over S which is a locally free module of finite type) and surjective (hence g is faithfully flat and quasi-compact, hence a descent morphism). Let X' be an S' -prescheme endowed with a descent datum. For this datum to be effective, it is necessary and sufficient that for every $x' \in X'$, the fiber $R(x') = q_2(q_1^{-1}(x'))$ be contained in an affine open set (a condition automatically satisfied if X' is quasi-projective over S').*

The remark in parentheses comes from the fact that if s is the point of S below x' , then $R(x')$ is finite and contained in the fiber X'_s ; on the other hand, since X' is quasi-projective over S' and S' is finite over S , X' is quasi-projective over S , which implies that a fiber of X' over S is contained in an affine open set.

Since every finite subset of an affine scheme admits a fundamental system of affine neighborhoods, one sees that one does not lose the hypothesis by restricting oneself to the part over an affine open set of S , which by virtue of 7.3 reduces us to the case where S is affine. By virtue of 7.4, we are reduced to showing that x' is contained in an affine open set *stable* under the descent datum. Indeed, let U be an affine open set containing $R(x')$; then the saturation

$$R(X' - U) = q_2(q_1^{-1}(X' - U))$$

does not meet $R(x')$; on the other hand, since q_2 is finite (because g , hence p_2 , is) and therefore closed, the right-hand side is a closed subset of X' . Let U' be its complement in X' ; it is therefore a *saturated* open set and one has

$$R(x') \subset U' \subset U$$

with U affine, but U' not affine a priori. Since a finite subset $R(x')$ in an affine scheme U has a fundamental system of affine neighborhoods of the form U_f , one sees, replacing f by its restriction to U' , that there exists a section f of $\mathcal{O}_U$ such that:

$$R(x') \subset U'_f, \quad U'_f \text{ is affine.}$$

Then let $U'' = q_1^{-1}(U') = q_2^{-1}(U')$, let us again denote by q_1, q_2 the induced morphisms $U'' \rightarrow U'$, and consider

$$f' = N_{q_2}(q_1^*(f)),$$

where N_{q_2} denotes the *norm* relative to the finite locally free morphism $q_2: U'' \rightarrow U'$. The compatibility of the formation of the norm with change of base easily implies that f' is an *invariant* section:

$$q_1^*(f') = q_2^*(f')$$

which implies that $U'_{f'}$ is a saturated open set of U' . More precisely, in fact, denoting by $Z(f')$ the set of zeros of a section f' , one finds, by virtue of the properties of norms:

$$Z(f') = q_2(Z(q_1^*(f))) = q_2(q_1^*(Z(f))) = R(U' - U'_f),$$

which implies that $U'_{f'} = U' - Z(f')$ is saturated, contains $R(x')$, and is contained in U'_f . Since the latter is affine, it follows that $U'_{f'}$ is affine too (since it is equal to $(U'_f)_{f''}$, with $f'' = f'|_{U'_f}$). It is therefore a saturated affine open set containing $R(x')$, hence x' , which completes the proof.

One will note that this reasoning applies whenever one has an equivalence relation (or even only a pre-equivalence relation, cf. [3]) in a prescheme X' , finite and locally free, and moreover 7.6 is also a special case of the analogous result for finite and locally free pre-equivalences, cf. loc. cit. The same remark applies to 7.7 below.

One can also, once the existence of a saturated quasi-affine open set U' containing x' has been obtained, appeal to 7.9 and 7.2, which avoids recourse to norms.

Let us note moreover that under the conditions of 7.6, if the descent datum on X' is effective, X' coming from X over S , then the morphism $X' \rightarrow X$ is finite, locally free and surjective, since it is deduced from g by the change of base $X \rightarrow S$. It follows (EGA II 6.6.4) that if X' is quasi-projective over S' , hence over S , then X is quasi-projective over S (a relatively ample invertible sheaf on X being obtained by taking the *norm* of an invertible sheaf on X' relatively ample over S , or over S' , which amounts to the same). One thus obtains:

COROLLARY 7.7. *A finite locally free and surjective morphism $g: S' \rightarrow S$ is an effective descent morphism for the fibered category of preschemes quasi-projective over other ones,*

i.e. for every X' quasi-projective over S' , every descent datum on X' relative to g is effective, and the descended S -prescheme X is quasi-projective over S .

PROPOSITION 7.8. *Let $g: S' \rightarrow S$ be a faithfully flat and quasi-compact morphism. Then g is an effective descent morphism for the fibered category of preschemes Z quasi-compact over a prescheme T , endowed with an invertible sheaf ample relative to T . In particular, for every prescheme X' over S' endowed with a descent datum relative to $g: S' \rightarrow S$, and every invertible sheaf $\mathcal{L}'$ on X' ample relative to S' , also endowed with a descent datum relative to the one given on X' (i.e. endowed with an isomorphism of $q_1^*(\mathcal{L}')$ with $q_2^*(\mathcal{L}')$ satisfying the usual transitivity condition), the descent datum on X' is effective, and the invertible sheaf $\mathcal{L}$ on the descended prescheme X , deduced from $\mathcal{L}'$ by descent, is ample relative to S .*

The proof is quite analogous to that of 5.8, noting that on the quasi-coherent graded Algebra $\mathcal{S}'$ over S' defined by $\mathcal{L}'$ there is a descent datum, making it possible to construct a quasi-coherent graded Algebra $\mathcal{S}$ over S thanks to 1.1, whence a $P = \text{Proj}(\mathcal{S})$ over S such that $P' = \text{Proj}(\mathcal{S}')$ is identified, together with its descent datum, with $P \times_S S'$. Since by hypothesis X' is identified with an open set of P' , necessarily stable under the descent datum on P' , the descent datum on X' is also effective, and one obtains the descended prescheme as an open set in P . The details are left to the reader. — In particular, taking $\mathcal{L}' = \mathcal{O}_{X'}$, one finds:

COROLLARY 7.9. *Let $g: S' \rightarrow S$ be a faithfully flat and quasi-compact morphism, and let X' be a quasi-affine prescheme over S' ; then every descent datum on X' relative to g is effective, and the descended prescheme X is quasi-affine over S .*

By virtue of 6.2, this result applies in particular if S' is locally noetherian and X' is quasi-finite and separated over S' , more generally if S' is arbitrary and X' is of finite presentation, quasi-finite and separated over S' (cf. 6.6).

REMARKS 7.10. The results given in this no. exhaust the effectiveness criteria currently known, and probably even the useful criteria that exist⁶. One will note the following counterexamples in support of this assertion:

- (i) If S is the spectrum of a field, and S' is the spectrum of a Galois quadratic extension, one can find an X' over S' , proper and smooth over S' , of dimension 3, endowed with a descent datum which is not effective (Serre).
- (ii) One can find an S spectrum of a regular local ring of dimension 3 (if one wishes, the local ring of an algebraic scheme over a field of given characteristic), a T principal covering of S with group $\mathbf{Z}/2\mathbf{Z}$, such that, if t denotes one of the points of T above the closed point s of S , and $S' = T - s$, one can find an X' projective over S' , regular, endowed with a descent datum relative to $g: S' \rightarrow S$, this descent datum not being effective.

For these constructions one uses Hironaka's example of non-projective varieties. For (i), it suffices to use the fact that one can find over k a proper and smooth scheme X_0 of dimension 3, on which $G = \mathbf{Z}/2\mathbf{Z}$ acts without inertia, and in which there exist two points a, b rational over S , congruent under G , which are not contained in an affine open set. One then puts $X' = X_0 \times_k k'$, and one lets G act on X' by means of the actions of G on the two factors, which gives a descent datum on X' relative to $g: \text{Spec}(k') \rightarrow \text{Spec}(k)$. Above a resp. b , there is exactly one point a' resp. b' (with quadratic residue field extension), and a' and b' are congruent under G , since $X' \rightarrow X_0$ is compatible with the actions of G . Then a' and b' cannot be contained in an affine open set, say U' , for then $U = X_0 - \text{Im}(X' - U')$ would be an open set of X_0 containing (a, b) and whose inverse image in X' would be contained

⁶This opinion has turned out to be partially mistaken, see for example J.P. MURRE, Sémin. Bourbaki 294 (Appendix), May 1965, and special results (notably of NÉRON and RAYNAUD). For the descent of group schemes; cf. M. Raynaud, thesis (1968).

in U' , hence quasi-affine, hence U would be quasi-affine, and consequently (a, b) would have an affine neighborhood in U .

For (ii), one uses the fact that in Hironaka's example, X_0 is obtained as a prescheme proper over a projective k -scheme Y , smooth over k (the morphism $f: X_0 \rightarrow Y$ being moreover birational, but that does not matter), the group G also acting on Y compatibly with its actions on X_0 ; finally, putting $S' = Y - f(b)$, $X' = X_0|_{S'}$, X' is projective over S' . Then X_0 is endowed with a natural descent datum relative to the canonical morphism $Y \rightarrow S = Y/G$, thanks to the actions of G on X_0 compatible with its actions on Y . This descent datum is not effective, since (a, b) is not contained in an affine open set. The induced descent datum on X' relative to $g: S' \rightarrow S$ is then not effective, as one easily verifies.

8. Bibliography

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